RVMC Clinical Operations

GPC Clinical Corner

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Antibiotics for Sinusitis

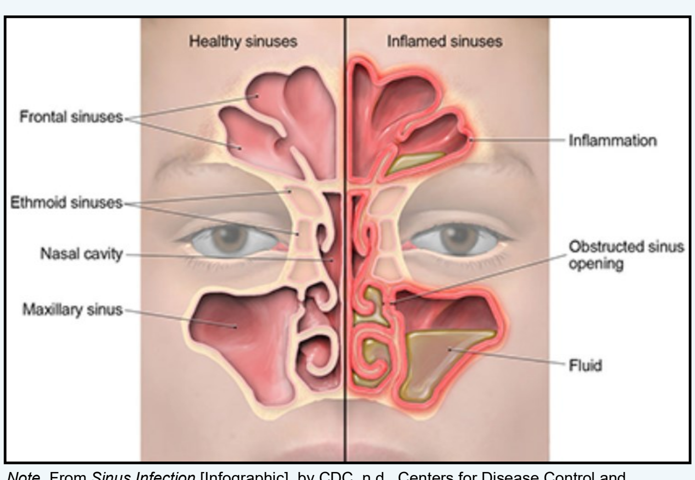
Recommendation

Don't routinely prescribe antibiotics for acute mild to moderate sinusitis unless symptoms last for 10+ days **or** symptoms worsen after initial clinical improvement. (Adult sinusitis symptoms must include discolored nasal secretions **and** facial or dental tenderness to percussion. Please be aware that children with viral URIs may often complain of discolored nasal secretions.)

Most sinusitis in the ambulatory setting is due to a viral infection that will resolve on its own. Despite consistent recommendations to the contrary, antibiotics are prescribed in over 80% of outpatient visits for acute sinusitis.

Supporting Information

Why Avoid Prescribing Antibiotics Immediately?



Note. From *Sinus Infection* [Infographic], by CDC, n.d., Centers for Disease Control and Prevention (https://www.cdc.gov/antibiotic-use/sinus-infection.html)

Sinusitis is one of the most common diagnoses in primary care. **Current recommendations strongly support not prescribing antibiotics within the first week of illness for mild to moderate sinusitis.** A meta-analysis published in 2012 in the Archives of Internal Medicine states that some randomized controlled trials showed that the patients assigned to antibiotics had a 7% to 14% higher rate of improvement in symptoms. However, researchers concluded that the potential harms from using antibiotics to manage sinusitis, including adverse effects (e.g., diarrhea), increased antibiotic resistance, and cost, clearly outweigh the potential minor benefits.

A Cochrane review compiled data from 59 studies that involved the use of antibiotics to manage simple maxillary sinus infection in primary care settings. **Studies that compared antibiotics with placebo showed that, in most cases, symptoms improved within two weeks, regardless of whether the participant received antibiotics**. The review found that, in addition to patient-related adverse effects (e.g., skin rash, abdominal pain, vomiting), antibiotic use poses the risk of increased resistance to antibiotics among community-acquired pathogens.

Symptomatic Care Options for Adults with Acute Rhinosinusitis

- **Analgesics and antipyretics:** Over-the-counter analgesics and antipyretics, such as nonsteroidal anti-inflammatory drugs (NSAIDs) and acetaminophen, can be used for pain and fever relief.

Avoid/Limit	
Acetaminophen	Advanced liver disease or decompensated cirrhosis
NSAIDs	Cardiovascular disease, chronic kidney disease, or advanced liver disease or cirrhosis. NSAIDs can be associated with increased risk of bleeding and gastrointestinal upset.

Note. From *Neti-Pot* [Photograph], by FDA, n.d., U.S. Food and Drug Administration (https://www.fda.gov/consumers/consumer-updates/rinsing-your-sinuses-neti-pots-safe)

- **Saline irrigation:** Mechanical irrigation with buffered, physiologic, or hypertonic saline may reduce the need for pain medication and improve overall patient comfort, particularly in patients with frequent sinus infections. The evidence supporting the use of saline irrigation is limited but indicates possible benefits for symptom relief with minor adverse effects, such as nasal burning and irritation. It is important that sinus rinses be prepared from sterile or bottled water, as there have been reports of amebic encephalitis due to tap water rinses. Sinus rinse kits are available for patients 2 years and older.

- **Intranasal glucocorticoids**: Studies have shown small symptomatic benefits and minimal adverse effects with short-term use of intranasal glucocorticoids for adult patients with both viral and bacterial ARS. Intranasal glucocorticoids are likely to be most beneficial for patients with underlying allergic rhinitis. The theoretical mechanism of action is a decrease in mucosal inflammation that allows improved sinus drainage.

Class	Selected Medication Options	Advantages	Disadvantages
Intranasal glucocorticoid sprays (preferred)	• fluticasone propionate • mometasone • triamcinolone	• Relieve congestion by reducing inflammation • May be particularly helpful for patients with allergic rhinitis	• Can cause epistaxis and sore throat
Intranasal saline spray	• Intranasal sterile saline	• Moisturizes passages and loosens secretions • May temporarily improve nasal passage patency • Useful combined with intranasal glucocorticoid sprays	• Some patients may find this difficult or uncomfortable • Saline must be sterile
Intranasal anticholinergic spray	• ipratropium bromide	• Significantly reduces rhinorrhea	• May not improve congestion
Intranasal decongestant sprays	• oxymetazoline	• Can improve nasal potency and promote drainage	• May cause rebound congestion or mucosal damage when used for long periods • Shouldn't be used for >3 days
Oral decongestants	• pseudoephedrine • phenylephrine	• Relieves congestion through vasoconstriction • Pseudoephedrine may be more effective than phenylephrine • May be particularly helpful for patients with Eustachian tube dysfunction (e.g., ear pain, a sensation of ear fullness or pressure, hearing loss, and/or tinnitus)	• Avoid or use with caution in patients with cardiovascular disease, hypertension, angle-closure glaucoma, or bladder neck obstruction due to sympathomimetic side effects
Oral antihistamines	**First generation** • clemastine • diphendydramine **Second generation** • fexofenadine • loratadine	• First-generation agents can be useful for drying effect • Second-generation agents can be useful in patients with allergies • Available in combination with oral decongestants	• Can lead to over-drying; thickened, difficult-to-mobilize secretions, and increased discomfort • Can cause drowsiness, cognitive impairment, and anticholinergic effects

- **Antibiotics**: For patients who do not have good follow-up, we initiate antibiotic therapy upon an ABRS (acute bacterial rhinosinusitis) diagnosis. In addition, antibiotics should also be started in patients who have been managed with observation, who have worsening symptoms or fail to improve within a seven-day period after clinician diagnosis. For more details on pediatric sinusitis, please visit the May 25, 2023 edition of GPC Clinical Corner.

  Treatment decisions for immunocompromised patients should be made on a case-by-case basis. Such patients may warrant immediate antibiotic treatment.

Adult Bacterial Rhinosinusitis		
No risk factors for pneumococcal resistance	amoxicillin	500 mg TID or 875 mg BID
	amoxicillin-clavulanate	500/125 mg TID or 875/125 mg BID
	doxycycline (PCN allergy)	100 mg BID or 200 mg QD
Risk factors for pneumococcal resistance*	levofloxacin (PCN allergy)**	500 mg or 750 mg QD
	moxifloxacin (PCN allergy)**	400 mg QD

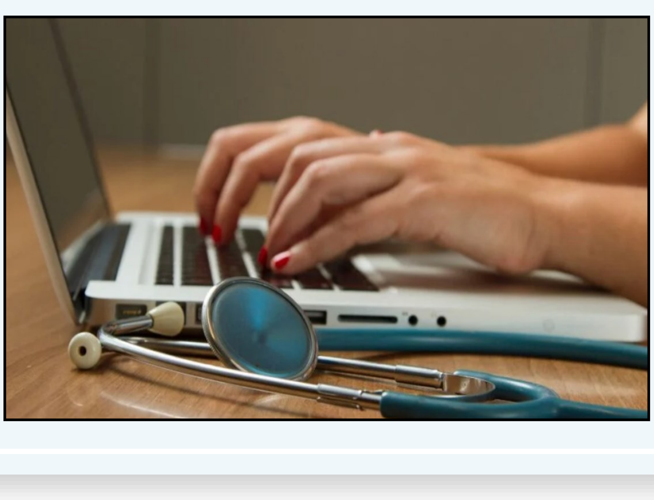
*The addition of clindamycin to a cephalosporin provides improved coverage for beta-lactam-resistant *S. pneumoniae*, although this carries an increased risk of adverse effects (e.g., *Clostridioides* [formerly *Clostridium*] *difficile* infection).

**Patients who fail ≥2 courses of appropriate antibiotics should have imaging and be referred for further evaluation.

Pediatric Acute Bacterial Rhinosinusitis		
First-line treatment	amoxicillin	90 mg/kg/day PO ÷ BID x 10 days (max 1000 mg/dose)
Second-line treatment	amoxicillin-clavulanate ES-600	90 mg/kg/day PO ÷ BID x 10 days (max 1000 mg/dose)
	clindamycin + 2nd/3rd-gen. cyclosporin (cephalosporin)*	30 mg/kg/day PO ÷ TID (max 600 mg/dose) + *see asterisk
	doxycycline (PCN allergy)	100 mg PO BID x 10 days
	levofloxacin (PCN allergy)	**6 mos to 5 yrs**: 16 to 20 mg/kg/day divided every 12 hrs **5 to 17 yrs**: 8 to 10 mg/kg/day every 24 hrs

** cefdinir or cefpodoxime. May use cefuroxime tabs for older children.*

Introducing the GPC Mailbag!



At GPC Clinical Corner, we strive to provide tips on the best clinical practices, navigating patient encounters, and updated information on treatment options.

This year, we'd like to expand our scope to include your suggestions. We're proud to announce the GPC Mailbag, where you can request topics for discussion, or even specific questions, that you'd like to see in a future newsletter. You can reach us at **GPCMailbag@kp.org**. We look forward to hearing from you!

References

Centers for Disease Control and Prevention (n.d.). *Sinus Infection* [Infographic]. Centers for Disease Control and Prevention. https://www.cdc.gov/antibiotic-use/sinus-infection.html

Kaiser Permanente. (n.d.) Adult antibiotic gold card. Kaiser Permanente. https://cl.kp.org/scal/home/refcontainerpage.html/content/clinicallibrary/ncal/clib/guidelines/infection_control/ca-empiric-id-guidelines-adults.nohf.ref.html?q=ANTIBIOTIC%20GOLD%20CARD&context=searchkp

Kaiser Permanente. (n.d.) Pediatric antibiotic gold card. Kaiser Permanente. https://cl.kp.org/scal/home/refcontainerpage.html/content/clinicallibrary/ncal/clib/guidelines/infection_control/ca-empiric-id-guidelines-peds.nohf.ref.html?q=pediatric%20gold%20card#21

U.S. Food and Drug Administration. (n.d.) *Neti-Pot* [Photograph]. U.S. Food and Drug Administration. https://www.fda.gov/consumers/consumer-updates/rinsing-your-sinuses-neti-pots-safe